8 Projects (optional, for fun)

QUICK

Project 7. Automatic watering, safely

Quick version

No time to read it all? This is the short version. The full guide is below.

- A low-voltage valve (or small pump) waters a bed for a set time when the soil is dry, skips watering when rain is coming, and stops with one button.
- Medium to hard, about 3 hours, about $35 to $60. Do Projects 3, 5 and 6 first.
- Low voltage only: a 12 V adapter and 5 V USB. If a part plugs into a wall outlet or says 110 V, 120 V or 230 V, do not use it.
- Use a **normally closed** valve: if power or Wi-Fi fails, the water stops.
- The ESP32, the relay and every wire joint go inside the waterproof box. Only the valve gets wet.
- Do the whole project dry first, with the water OFF. Ask a friend who knows electrics to check the wiring before you first plug it in.
- The board itself closes the valve after 10 minutes, whatever Home Assistant says. Add a `STOP WATER` button to your dashboard too.

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